

THE THEORY OF MEASURES

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ABSTRACT. This is an investigation and introduction of the theory of measures. Covering length, area, volume, and probabilities, measures also form the basis of what integration means.

1. MOTIVATION

Length, Area, and Volume

We start with a simple question of *what is length?* We should be able to get some mathematical model for it. The standard objects of length measurements are one-dimensional, so it is fair to restrict ourselves to working in $\mathbb{R}$. In this setting, the objects of interest are, at their simplest, line segments, which are in turn naturally modeled by *intervals*.

So if we have an interval $[a, b] \subseteq \mathbb{R}$, it is then quite immediate to suggest its length:

$$\text{length of } [a, b] = b - a$$

2. BASIC DEFINITIONS AND RESULTS

Definition 2.1. Given a set Ω , a σ -algebra of subsets of Ω is a family $\mathcal{F} \subseteq \mathcal{P}(\Omega)$ such that

- (i) $\emptyset, \Omega \in \mathcal{F}$,
- (ii) if $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$, and
- (iii) if $\{A_k\}_{k \in K} \subseteq \mathcal{F}$ then $\bigcup_{k \in K} A_k \in \mathcal{F}$.

We'll say a pair $(\Omega, \mathcal{F})$ is a *measurable space*.

So σ -algebras are closed under countable unions, but they are also closed under countable intersections!

Question 1. Show that if $\{A_k\}_{k \in K} \subseteq \mathcal{F}$ then

$$\bigcap_{k \in K} A_k \in \mathcal{F}.$$

Let's pause here for a moment to ask, what kinds of σ -algebras are there?

Example 2.2. If Ω is a set, then $\mathcal{P}(\Omega)$ is a σ -algebra.

Actually *all* finite σ -algebras are built from these. (See Question 9)

But it would be dull if all σ -algebras were just power sets.

Definition 2.3. Given a set Ω and a family of subsets $\mathcal{S} \subseteq \mathcal{P}(\Omega)$, we define the σ -algebra generated by $\mathcal{S}$ to be the smallest σ -algebra containing $\mathcal{S}$. That is,

$$\sigma(\mathcal{S}) := \bigcap_{\substack{\mathcal{F} \subseteq \mathcal{P}(\Omega) \\ \mathcal{F} \text{ is a } \sigma\text{-algebra}}} \mathcal{F}.$$

It is a simple exercise to show that these generated σ -algebras are indeed σ -algebras. But how are these useful? Well one example of when we deal with a set of subsets that is not inherently a σ -algebra, is the setting of a topological space.

Example 2.4. Let (X, τ) be a topological space. Then the σ -algebra generated by the topology (i.e. all the open sets), *Borel σ -algebra*, which we will denote by

$$\mathcal{B}(X, \tau) := \sigma(\tau).$$

For example, take $\mathbb{R}$ with its usual (metric) topology. Then the Borel σ -algebra is a non-trivial (i.e. not $\mathcal{P}(\mathbb{R})$) σ -algebra containing all open subsets of $\mathbb{R}$. We will revisit this non-trivialness after introducing the Lebesgue measure to show, using the Axiom of Choice to construct “Vitali sets”, which are non-Borel subsets of $\mathbb{R}$.

Question 2. Show that $\mathcal{B}(\mathbb{R})$ contains all closed sets, all F_σ sets, and all G_δ sets.

Curiosity 1. What is the cardinality of $\mathcal{B}(\mathbb{R})$? Seems like it contains a lot of sets, so it could be that $|\mathcal{B}(\mathbb{R})| = |\mathcal{P}(\mathbb{R})| = 2^{\mathfrak{c}}$.

However, it definitely does not contain all sets. So perhaps we have $|\mathcal{B}(\mathbb{R})| = \mathfrak{c}$?

If the latter is the case, that would be quite shocking, as all imaginable or constructable sets appear to be Borel, yet there is a whole size of infinity larger than the Borel sets composed of the unimaginable ones!

Question 3. Show that the Borel subsets of $\mathbb{R}$ (with its metric topology) are also generated by

- (i) the half open intervals: $\{(a, b] : -\infty \leq a < b < \infty\}$,
- (ii)

In fact, we have the following. The proof is a challenging exercise, but not necessary to get stuck on.

Fact 1. If $\mathcal{F}$ is an infinite σ algebra, then $\mathcal{F}$ is uncountable.

Definition 2.5. Given a measurable space $(\Omega, \mathcal{F})$, a (positive) *measure* on $\mathcal{F}$ (or on Ω if $\mathcal{F}$ is understood) is a function $\mu : \mathcal{F} \rightarrow [0, \infty]$ such that

- (i) $\mu(\emptyset) = 0$, and
- (ii) if $\{A_k\}_{k \in K} \subseteq \mathcal{F}$ with $A_i \cap A_j = \emptyset$ for all $i \neq j$, then

$$\mu \left(\bigcup_{k=1}^{\infty} A_k \right) = \sum_{k=1}^{\infty} \mu(A_k).$$

A triple $(\Omega, \mathcal{F}, \mu)$ where $\mathcal{F}$ is a σ -algebra of subsets of Ω and μ is a measure on $\mathcal{F}$ will be called a *measure space*.

Our notions of length, area, and probability are all examples of measures, but their formal constructions will have to wait until we have some more tools. But let's see a simple but elegant example that will end up bridging the gap between sums and integrals.

Example 2.6. Take $\Omega = \mathbb{N}$ and $\mathcal{F} = \mathcal{P}(\mathbb{N})$. Define the *counting measure* on $\mathbb{N}$, $\mu_{\text{count}} : \mathcal{P}(\mathbb{N}) \rightarrow [0, \infty]$ by

$$\mu_{\text{count}}(S) = |S|.$$

It is not hard to see that this satisfies all requirements of our definition.

In fact, we can define this same counting measure on any set, which we may encounter later. But the counting measure on $\mathbb{N}$ will be of particular importance to us.

One may reasonably call this a zero-dimensional volume measure, like length is to one-dimension and area is to two-dimensions!

Example 2.7. More fundamentally, for any Ω and any $a \in \Omega$, we have the *Dirac mass at a* measure, which is defined by

$$\delta_a(S) = \begin{cases} 1 & \text{if } a \in S \\ 0 & \text{otherwise} \end{cases}.$$

If measures are showing us how volume or mass is distributed, then this one says that all of it is concentrated at the point a . It is a point-mass!

Dirac masses are the simplest measures we can have but they are surprisingly useful.

Question 4. *The two measures we saw above are connected in a very direct way. Show that*

$$\mu_{count} = \sum_{n=1}^{\infty} \delta_n.$$

Question 5. *Motivated from the previous question, show that adding two measures results in a measure.*

Not all linear combinations of measures are permissible. Which ones are?

Such a space is often called a convex cone.

We also have measures associated to length and area, as we started out. But showing their existence is a bit more involved. For now, we delay this problem to a future section.

The requirement for the additivity of measure that we have disjoint sets is not a big limitation, for we can always *disjointify* a family, and at least get sub-additivity.

Question 6. *For a given family of sets $\{A_k\}_{k \in K}$, show that there is a disjoint family $\{B_k\}_{k \in K}$ such that*

$$\bigcup_{k=1}^n A_k = \bigcup_{k=1}^n B_k.$$

Question 7. *Show that, with the setup above,*

$$\mu \left(\bigcup_{k=1}^{\infty} A_k \right) = \sum_{k=1}^{\infty} \mu(B_k).$$

Argue that for any countable collection $\{A_k\}_{k \in K} \subseteq \mathcal{F}$, we have sub-additivity of measure:

$$\mu \left(\bigcup_{k=1}^{\infty} A_k \right) \leq \sum_{k=1}^{\infty} \mu(A_k).$$

We also quickly get using the same ideas the following computational tool for dealing with non-disjoint sets:

Theorem 2.8 (Inclusion-Exclusion). *Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and let $A, B \in \mathcal{F}$. Then*

$$\mu(A \cup B) = \mu(A) + \mu(B) - \mu(A \cap B).$$

Remark 2.9. One other objection one might have to the definition of measure is that we can seemingly define it equally fine for just finite unions. For example, one can define a *finitely-additive measure* on an algebra $\mathcal{F}$ of subsets of Ω as a map $\mu : \mathcal{F} \rightarrow [0, \infty]$ satisfying

- (i) $\mu(\emptyset) = 0$, and
- (ii) If $A_1, \dots, A_n \in \mathcal{F}$ are pairwise disjoint, then

$$\mu(A_1 \cup \dots \cup A_n) = \mu(A_1) + \dots + \mu(A_n).$$

It is immediate that all measures in our original sense are finitely additive in this sense (why?). So, what benefit do we get from allowing for infinite unions?

A short possible justification is that by allowing countably infinities, we can deal with notions of limits and continuity, which vastly expands our scope. These kinds of arguments will be crucial for developing Lebesgue's integral later on.

The following is a result that is unique to countably-additive measures.

Theorem 2.10 (Continuity of Measure). *If $\{A_k\}_{k \in \mathbb{N}} \subseteq \mathcal{F}$ with $A_k \subseteq A_{k+1}$ for all $k \in \mathbb{N}$ then*

$$\mu \left(\bigcup_{k=1}^{\infty} A_k \right) = \lim_{k \rightarrow \infty} \mu(A_k).$$

Furthermore, if $\{A_k\}_{k \in \mathbb{N}} \subseteq \mathcal{F}$ with $A_k \supseteq A_{k+1}$ for all $k \in \mathbb{N}$, and if $\mu(A_1) < \infty$, then

$$\mu \left(\bigcap_{k=1}^{\infty} A_k \right) = \lim_{k \rightarrow \infty} \mu(A_k).$$

Question 8. Show that without the assumption that $\mu(A_1) < \infty$ in the second part of the theorem above, the statement may not be true.

2.1. Exercises.

Question 9. We will show that all finite measurable spaces are of the form $(\Omega, \mathcal{P}(\Omega'))$. Let $(\Omega, \mathcal{F})$ be a measurable space where $|\mathcal{F}| < \infty$.

- i Say a nonempty subset $A \in \mathcal{F}$ is minimal if $B \in \mathcal{F}$ and $B \subseteq A$ implies that $B = \emptyset$ or $B = A$. Show that the collection of minimal elements of $\mathcal{F}$ partitions Ω (i.e. they are all disjoint and union to Ω).
- ii Find an appropriate set Ω' and a function $f : \Omega \rightarrow \Omega'$ such that $\hat{f} : \mathcal{F} \rightarrow \mathcal{P}(\Omega')$, defined by $\hat{f}(A) := f(A)$ is an order-isomorphism of $\mathcal{F}$ to $\mathcal{P}(\Omega')$.

Question 10. Prove Fact 1. (Hint: $|\mathcal{P}(\mathbb{N})| > |\mathbb{N}|$)

Question 11. Show that the σ -algebra generated by singletons on $\mathbb{R}$ is not $\mathcal{P}(\mathbb{R})$.

Question 12. Show that there are finitely-additive measures that are not countably-additive. (Hint: such a measure must fail the continuity theorem for some collection of sets)

3. LEBESGUE'S INTEGRAL

4. THE RIESZ-MARKOV-KAKUTANI REPRESENTATION THEOREM